

Research Plan: AdaDA-TransUNet

1 Title

AdaDA-TransUNet: Hardware-Aware Adaptive Dual Attention Transformer UNet for Efficient and Precise Medical Image Segmentation

2 Dual Attention Network (DANet)

DANet [1] 由 Fu et al. 提出, 用于自然场景图像分割, 核心在于同时建模空间依赖 (位置 attention) 和通道依赖 (channel attention), 通过两种 self-attention 模块协同增强语义一致性和判别性:

2.1 位置注意力模块 (PAM: Position Attention Module)

- 输入特征图 $A \in R^{C \times H \times W}$
- 经卷积映射生成 $B, C \in R^{C \times H \times W}$, reshape 成 $R^{C \times N}$ 其中 $N = H \times W$
- 构建空间注意力图:

$$S_{ji} = \frac{\exp(B_i \cdot C_j)}{\sum_{i=1}^N \exp(B_i \cdot C_j)}$$

- 最终输出:

$$E_j = \alpha \sum_{i=1}^N S_{ji} D_i + A_j$$

- 具备全局感知能力, 适用于建模图像中“相似区域”的关联性 (即使相距较远)

2.2 通道注意力模块 (CAM: Channel Attention Module)

- 输入特征图 $A \in R^{C \times H \times W}$, reshape 成 $A^{C \times N}$
- 构建通道注意力图:

$$X_{ji} = \frac{\exp(A_i \cdot A_j)}{\sum_{i=1}^C \exp(A_i \cdot A_j)}$$

- 输出:

$$E_j = \beta \sum_{i=1}^C X_{ji} A_i + A_j$$

- 目标是捕捉各语义通道之间的互相关联, 提升通道表达能力

3 DA-TransUNet and Limitations

DA-TransUNet [2, 3] incorporates DA blocks into a Transformer U-Net. Its limitations:

- High complexity (PAM $\mathcal{O}(N^2)$, CAM $\mathcal{O}(C^2)$)
- Fixed compression ratio
- No hardware-aware inference adaptation

4 Our Optimizations

AdaDA-TransUNet improves DA-TransUNet via:

4.1 低秩+局部感知优化 (Low-rank Swin-style PAM) (降低 $\mathcal{O}(N^2)$)

- 原始 DANet 的 PAM 为全局 self-attention, 复杂度为 $\mathcal{O}(N^2)$
- 改进: 采用 Swin-style窗口attention[4]+低秩分解 (如 $A \approx PQ$)
- 计算变为: $\mathcal{O}(N^2) \rightarrow \mathcal{O}(Nr)$

4.2 Group Attention优化CAM (降低 $\mathcal{O}(C^2)$)

- DANet 中 CAM 为全通道交互, 成本 $\mathcal{O}(C^2)$
- 改进: 将通道分成 G 个组, 每组独立做 attention (适用于医学中多类通道表达)
- 成本降为 $\mathcal{O}(C^2/G)$

4.3 通道压缩比可学习化 Learnable Compression Ratio

- DANet 固定通道压缩比为 1/4 (用于 PAM、CAM 计算)
- 改进: 加入 Learnable Ratio

$$r = \sigma(W \cdot F)$$

根据全局特征自动调整通道压缩程度, 增强适应性和硬件友好性

4.4 加入 Hardware-Aware 配置模块(推理可配置)

- 根据推理设备可用内存动态调整window size/rank/group数量

5 Architecture Overview

- Encoder: Conv $\rightarrow$ AdaDA Block $\rightarrow$ Transformer
- Skip Connections: With DA Blocks
- Decoder: Upsampling $\rightarrow$ Fusion $\rightarrow$ Segmentation Head

6 Methodology

6.1 Low-Rank PAM

$$A \approx PQ, \quad P \in \mathbb{R}^{N \times r}, \quad Q \in \mathbb{R}^{r \times N}$$

6.2 Grouped CAM

$$X_{ji}^{(g)} = \frac{\exp(A_i \cdot A_j)}{\sum_{k=1}^{C/G} \exp(A_i \cdot A_k)}$$

6.3 Learnable Channel Ratio

$$r = \sigma(W \cdot F), \quad W \in \mathbb{R}^{d \times 1}$$

6.4 Hardware-Aware Configuration

Listing 1: Runtime configuration

```
def hardware_config(free_mem):
    if free_mem > 8:
        return {"rank": 64, "window": 14, "groups": 4}
    elif free_mem > 4:
        return {"rank": 32, "window": 7, "groups": 8}
    else:
        return {"rank": 16, "window": 7, "groups": 16}
```

6.5 DA Block Pseudocode

Listing 2: AdaDA Block (PyTorch-style)

```
class AdaDABlock(nn.Module):
    def __init__(self, in_channels, reduction=16, groups=8):
        super().__init__()
        self.pool = nn.AdaptiveAvgPool2d(1)
        self.ratio_fc = nn.Linear(in_channels, 1)
        self.groups = groups
        self.reduce = nn.Conv2d(in_channels, in_channels // reduction, kernel_size=1)
        self.restore = nn.Conv2d(in_channels // reduction, in_channels, kernel_size=1)
        self.group_cam = nn.Conv2d(in_channels, in_channels, kernel_size=1, groups=groups)

    def forward(self, x):
        B, C, H, W = x.shape
        global_feat = self.pool(x).view(B, -1)
        ratio = torch.sigmoid(self.ratio_fc(global_feat)).clamp(0.0625, 0.5)
        ch = int(C * ratio.item())
        x_reduced = x[:, :ch, :, :]
        cam_out = self.group_cam(x)
        fused = self.restore(x_reduced) + cam_out
        return fused
```

7 Loss Function

$$\mathcal{L} = \frac{1}{2} \cdot \mathcal{L}_{\text{Dice}} + \frac{1}{2} \cdot \mathcal{L}_{\text{CE}}$$

8 Experiment Plan

8.1 Datasets

- Synapse
- ISIC2018
- Chest X-ray
- CVC-ClinicDB
- Kvasir-Seg

8.2 Metrics

Dice/IoU/HD95/Params/FLOPs/FPS

8.3 Ablation Studies

- PAM: global vs. windowed vs. low-rank
- CAM: full vs. grouped
- Ratio: fixed vs. learnable
- With/without hardware config

9 Contributions

- First adaptive DA Transformer UNet with hardware-aware configuration
- Achieves better accuracy-efficiency trade-off
- Plug-and-play modules for modern segmentation architectures

References

- [1] J. Fu, J. Liu, H. Tian, Y. Li, Y. Bao, Z. Fang, and H. Lu, “Dual attention network for scene segmentation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 3146–3154.
- [2] G. Sun, Y. Pan, W. Kong, Z. Xu, J. Ma, T. Racharak, L.-M. Nguyen, and J. Xin, “Da-transunet: integrating spatial and channel dual attention with transformer u-net for medical image segmentation,” *Frontiers in Bioengineering and Biotechnology*, vol. 12, p. 1398237, 2024.
- [3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
- [4] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, “Swin transformer: Hierarchical vision transformer using shifted windows,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10 012–10 022.